
A Steady State Incompressible Fluid Flow Solver using the SIMPLE method

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Abstract

We present a code written in the Python programming language to solve the steady state incompressible transport equations over a uniform grid. Using the semi-implicit method for pressure-linked equations (SIMPLE) algorithm to solve for the velocity and relative pressure fields, and the finite volume method to solve for primitives like temperature, the code is robust to solving for many scenarios. Multiple discretization schemes are available to choose from (central difference, upwind, hybrid) and boundary conditions can be specified by the user. This paper includes solutions for problems such as the common lid-driven cavity and the back-step flow problem at different Reynolds numbers, successfully demonstrating the formation of vortices. The solutions are compared to the trusted references in the literature and show sufficient agreement. The code is written with ease-of-use in mind and partly optimized to maximally utilize array-based operations rather than loops to improve speed though there is more work to be done to further improve efficiency.

1 Introduction

The time-independent (steady state) transport of a property φ is given in conservative form

$$\nabla \cdot \rho \varphi \mathbf{u} = \Gamma \nabla \cdot \nabla \varphi - \nabla p + S_\varphi \quad (1)$$

where $\mathbf{u}$ is the velocity field, p the pressure field, ρ the mass density, Γ is the diffusion coefficient (dynamic viscosity in the case of the momentum equations) and S_φ the source term, which typically includes the pressure gradient though is separated from it in the SIMPLE method. φ is any transported primitive such as heat, or a velocity component in the Navier-Stokes momentum equations.

$$\nabla \cdot \rho \mathbf{u} = 0 \quad (2)$$

across the domain. In this paper we replace φ with u, v and Γ with the dynamic viscosity μ to yield the Navier-Stokes equations, compactly written as

$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = \mu \nabla \cdot \nabla \mathbf{u} - \nabla p + S_{\mathbf{u}} \quad (3)$$

$$\nabla \cdot \rho \mathbf{u} = 0 \quad (4)$$

2 Numerical Methodology

2.1 Non-dimensionalizing the Navier-Stokes Equations

To non-dimensionalize the Navier Stokes equations ($\varphi \in \{u, v\}$) we can define the characteristic length and speed L, U and set

$$\begin{aligned} \mathbf{x}' &:= \frac{\mathbf{x}}{L} \\ \mathbf{u}' &:= \frac{\mathbf{u}}{U} \\ p' &:= \frac{p}{\rho U^2} \end{aligned}$$

Noting that $\frac{\partial(\cdot)}{\partial x} = \frac{1}{L} \frac{\partial(\cdot)'}{\partial x'}$, $\frac{\partial(\cdot)}{\partial y} = \frac{1}{L} \frac{\partial(\cdot)'}{\partial y'}$ $\nabla_{(\cdot)} \mathbf{u} = U \nabla_{(\cdot)} \mathbf{u}'$ and the kinematic viscosity $\nu = \frac{\mu}{\rho}$ we have

$$\begin{aligned} \rho \frac{U^2}{L} \nabla_{\mathbf{x}'} \cdot (\mathbf{u}' \otimes \mathbf{u}') &= \mu \frac{U}{L^2} \nabla_{\mathbf{x}'} \cdot \nabla_{\mathbf{x}'} \mathbf{u}' - \frac{\rho U^2}{L} \nabla_{\mathbf{x}'} p' \\ &\equiv \nabla_{\mathbf{x}'} \cdot (\mathbf{u}' \otimes \mathbf{u}') = \frac{\mu}{\rho L U} \nabla_{\mathbf{x}'} \cdot \nabla_{\mathbf{x}'} \mathbf{u}' - \nabla_{\mathbf{x}'} p' \\ &\equiv \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = \frac{\nu}{L U} \nabla \cdot \nabla \mathbf{u} - \nabla p \end{aligned} \quad (5)$$

where in the final line we re-identify the variables without the '. We can now define the dimensionless Reynold's number as

$$Re := \frac{L U}{\nu} \quad (6)$$

and conclude

$$\nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = \frac{1}{Re} \nabla \cdot \nabla \mathbf{u} - \nabla p \quad (7)$$

2.2 Finite Volume Discretization

We now write the equilibrium equations for a given control volume and show how the SIMPLE method handles the coupling of pressure and velocity. We choose a collocated uniform grid, so that velocity and pressure are coincident with respect to nodes.

$$\nabla \cdot u \mathbf{u} = \frac{1}{Re} \nabla \cdot \nabla u - \frac{\partial p}{\partial x} \quad (8)$$

$$\nabla \cdot v \mathbf{u} = \frac{1}{Re} \nabla \cdot \nabla v - \frac{\partial p}{\partial y} \quad (9)$$

Integrating (8) over a 2D control volume

$$\begin{aligned} 0 &= \int_0^1 \int_{y_s}^{y_n} \int_{x_w}^{x_e} \frac{1}{Re} \nabla \cdot \nabla u - \frac{\partial p}{\partial x} - \nabla \cdot u \mathbf{u} dV \\ &= \int_A \left(\frac{1}{Re} \nabla u - p \hat{\mathbf{i}} - u \mathbf{u} \right) \cdot \mathbf{n} dS \\ &\approx \frac{1}{Re} \left(\left(\frac{\partial u}{\partial x} \right)_e dy - \left(\frac{\partial u}{\partial x} \right)_w dy + \left(\frac{\partial u}{\partial y} \right)_n dx - \left(\frac{\partial u}{\partial y} \right)_s dx \right) - (p_e - p_w) dy \\ &\quad - ((u u)_e dy - (u u)_w dy + (u v)_n dx - (u v)_s dx) \end{aligned} \quad (10)$$

and similarly for v

$$\begin{aligned} 0 &\approx \frac{1}{Re} \left(\left(\frac{\partial v}{\partial x} \right)_e dy - \left(\frac{\partial v}{\partial x} \right)_w dy + \left(\frac{\partial v}{\partial y} \right)_n dx - \left(\frac{\partial v}{\partial y} \right)_s dx \right) - (p_n - p_s) dx \\ &\quad - ((v u)_e dy - (v u)_w dy + (v v)_n dx - (v v)_s dx) \end{aligned} \quad (11)$$

Integrating (4) over a control volume yields

$$\begin{aligned} \int_0^1 \int_{y_s}^{y_n} \int_{x_w}^{x_e} \nabla \cdot \rho \mathbf{u} dV &= \int_A \rho \mathbf{u} \cdot \mathbf{n} dS \\ &\approx \rho dy (u_e - u_w) + \rho dx (v_n - v_s) \end{aligned} \quad (12)$$

Set the convective F and diffusive D

$$F_e = u_e, F_w = u_w, F_n = v_n, F_s = v_s \quad (13)$$

$$D_e = D_w = \frac{1}{Re \cdot dy}, D_n = D_s = \frac{1}{Re \cdot dx} \quad (14)$$

Firstly,

$$\begin{aligned} p_e - p_w &\approx \frac{p_E + p_P}{2} - \frac{p_W + p_P}{2} = \frac{p_E - p_W}{2} \\ p_n - p_s &\approx \frac{p_N - p_S}{2} \end{aligned}$$

so we aim to express (10) and (11) in the form

$$A_P u_P = \sum_{nb} A_{nb} u_{nb} - \frac{p_E - p_W}{2} dy + S_u \quad (15)$$

$$A_P v_P = \sum_{nb} A_{nb} v_{nb} - \frac{p_N - p_S}{2} dy + S_v \quad (16)$$

where nb refers to the east, west, north, and south neighbors and

$$A_P = \sum_{nb} A_{nb} + (F_e - F_w + F_n - F_s) - S_p \quad (17)$$

using a discretization scheme for the convection terms. The different schemes will now be summarized. Without loss of generality the **north** face will be used in the proceeding derivations of the terms for the u -momentum, though the procedure is identical for the other faces and the v -momentum discretized equation.

The terms involving u_N are

$$\frac{1}{Re} \left(\frac{\partial u}{\partial x} \right)_n dx \approx \frac{1}{Re} \frac{u_N - u_P}{dx} dx = D_n(u_N - u_P) dx \quad (18)$$

$$-(uv)_n dx = -F_n u_n dx \quad (19)$$

where (18) follows when u_N and u_P are a distance dx from each other; in the case where $u_n = u_{BN}$ is at a boundary, they are $\frac{dx}{2}$ apart and

$$\frac{1}{Re} \left(\frac{\partial u}{\partial x} \right)_n dx \approx 2D_n(u_{BN} - u_P) dx \quad (20)$$

2.2.1 The Central Difference Scheme

The central difference scheme sets

$$-F_n u_n dx = -F_n \frac{u_N + u_P}{2} dx \quad (21)$$

when u_n is unknown. We analyze three cases implemented in the code and see which terms contribute to A_N , A_P , S_u and S_p :

1. Interior cells:

$$A_N = (D_n - \frac{1}{2} F_n) dx$$

$$A_P \leftarrow A_P + A_N$$

2. Dirichlet Boundary: If the north face is a Dirichlet boundary $u = u_{BN}$, then

$$A_N = 0$$

$$S_p \leftarrow S_p + (2D_n - F_n) dx$$

$$S_u \leftarrow S_u + ((2D_n - F_n) dx) u_{BN}$$

3. Homogeneous Neumann Boundary: If the north face follows a homogeneous Neumann condition ($\nabla_n \mathbf{u} = \mathbf{0}$) then

$$A_N = 0$$

2.2.2 The Upwind Scheme

The upwind scheme sets u at a face to the upstream cell center value. That is

$$-F_n u_n dx = -F_n \left(\begin{cases} u_N & F_n < 0 \\ u_P & F_n \geq 0 \end{cases} \right) dx \quad (22)$$

1. Interior cells:

$$A_N = (D_n + \max(0, -F_n))dx$$

$$A_P \leftarrow A_P + A_N$$

2. Dirichlet Boundary:

$$A_N = 0$$

$$S_p \leftarrow S_p + (2D_n + \max(0, -F_n))dx$$

$$S_u \leftarrow S_u + ((2D_n + \max(0, -F_n))dx)u_{BN}$$

3. Homogeneous Neumann Boundary:

$$A_N = 0$$

2.2.3 The Hybrid Scheme

The hybrid scheme sets u depending on the peclet number $Pe = \frac{F_n}{D_n}$; the central difference scheme is used for low Pe whereas the upwind scheme without diffusion is used for high Pe :

$$-F_n u_n dx = -F_n \left(\begin{cases} \frac{u_N + u_P}{2} & |Pe| \leq 2 \\ u_N & F_n < 0, |Pe| > 2 \\ u_P & F_n > 0, |Pe| > 2 \end{cases} \right) dx \quad (23)$$

where the first case is only for unknown u_n as in (22), else the value is used.

1. Interior cells:

$$A_N = \max(D_n - \frac{1}{2}F_n, \max(0, -F_n))dx$$

$$A_P \leftarrow A_P + A_N$$

2. Dirichlet Boundary:

$$A_N = 0$$

$$S_p \leftarrow S_p + \max(2D_n - F_n, \max(0, -F_n))dx$$

$$S_u \leftarrow S_u + (\max(2D_n - F_n, \max(0, -F_n))dx)u_{BN}$$

3. Homogeneous Neumann Boundary:

$$A_N = 0$$

3 The SIMPLE Method

The SIMPLE method is used to iteratively solve the velocity and pressure fields taking into account their natural coupling given by (10). The algorithm works as follows:

1. Guess pressure field p^* and velocity fields u^* and v^* .

2. Solve the momentum equations:

The discretized momentum equation is

$$A_P u_P = \sum_{nb} A_{nb} u_{nb} - \frac{p_E - p_W}{2} dy + S_u \quad (24)$$

and with the tentative velocities and pressure:

$$A_P u_P^* = \sum_{nb} A_{nb} u_{nb}^* - \frac{p_E^* - p_W^*}{2} dy + S_u \quad (25)$$

Choose some $\alpha_u, \alpha_v, \alpha_p$ under-relaxation factors; here we choose $\alpha_u = \alpha_v = \alpha_m := 0.7$ and $\alpha_p = 0.3$. We define

$$u_P^{*(n)} := \alpha_u u_P^* + (1 - \alpha_u) u_P^{(n-1)} \quad (26)$$

$$\implies u_P^* = \frac{1}{\alpha_u} u_P^{*(n)} - \frac{(1 - \alpha_u)}{\alpha_u} u_P^{(n-1)} \quad (27)$$

$$(28)$$

and substituting into (25) gives

$$A_P \left(\frac{1}{\alpha_u} u_P^{*(n)} - \frac{(1 - \alpha_u)}{\alpha_u} u_P^{(n-1)} \right) = \sum_{nb} A_{nb} u_{nb}^* - \frac{p_E^* - p_W^*}{2} dy + S_u \quad (29)$$

$$\implies \frac{A_P}{\alpha_u} u_P^{*(n)} = \sum_{nb} A_{nb} u_{nb}^* - \frac{p_E^* - p_W^*}{2} dy + S_u + \frac{A_P(1 - \alpha_u)}{\alpha_u} u_P^{(n-1)} \quad (30)$$

For the v -momentum this is

$$\frac{A_P}{\alpha_v} v_P^{*(n)} = \sum_{nb} A_{nb} v_{nb}^* - \frac{p_E^* - p_W^*}{2} dy + S_v + \frac{A_P(1 - \alpha_v)}{\alpha_v} v_P^{(n-1)} \quad (31)$$

Solve (30) and (31) using a linear solver to produce the updated fields $u^{*(n)}, v^{*(n)}$. Henceforth the (n) is left out and velocities are understood as being derived after under-relaxing.

3. Use Rhie-Chow interpolation to calculate tentative face velocities. We show the interpolation for the east faces though the procedure is the same for the west face (for the y -momentum problem, the interpolation is for the north/south faces). Firstly, consider a cell P and its east neighbor cell E , both with zero sources. We rewrite (17) as follows for both cells

$$\frac{A_P|_P}{\alpha_u} u_P^*|_P = \sum_{nb} A_{nb}|_P u_{nb}^*|_P - \frac{p_E^* - p_W^*}{2} dy \quad (32)$$

$$\frac{A_P|_E}{\alpha_u} u_P^*|_E = \sum_{nb} A_{nb}|_E u_{nb}^*|_E - \frac{p_{EE}^* - p_P^*}{2} dy \quad (33)$$

We assume from the above a fictitious discretization at the east face

$$\frac{A_P|_e}{\alpha_u} u_P^*|_e = \sum_{nb} A_{nb}|_e u_{nb}^*|_e - (p_E^* - p_P^*) dy \quad (34)$$

$$\implies u_P^*|_e = \frac{\alpha_u}{A_P|_e} \sum_{nb} A_{nb}|_e u_{nb}^*|_e - \frac{\alpha_u}{A_P|_e} (p_E^* - p_P^*) dy \quad (35)$$

(35) is fictitious because the coefficients $A_{(\cdot)}$ do not actually exist, so Rhie-Chow [2] proposes to use the following interpolation:

$$\begin{aligned} \frac{1}{A_P|_e} \sum_{nb} A_{nb}|_e u_{nb}^*|_e &:= \frac{1}{2} \left(\frac{1}{A_P|_P} \sum_{nb} A_{nb}|_P u_{nb}^*|_P + \frac{1}{A_P|_E} \sum_{nb} A_{nb}|_E u_{nb}^*|_E \right) \\ &= \frac{1}{2} \left(\frac{u_P^*|_P}{\alpha_u} + \frac{p_E^* - p_W^*}{2A_P|_P} dy + \frac{u_P^*|_E}{\alpha_u} + \frac{p_{EE}^* - p_P^*}{2A_P|_E} dy \right) \end{aligned} \quad (36)$$

$$\frac{1}{A_P|_e} := \frac{1}{2} \left(\frac{1}{A_P|_P} + \frac{1}{A_P|_E} \right) \quad (37)$$

Substituting into (35) gives

$$\begin{aligned} u_P^*|_e &= \frac{\alpha_u}{2} \left(\frac{u_P^*|_P}{\alpha_u} + \frac{p_E^* - p_W^*}{2A_P|_P} dy + \frac{u_P^*|_E}{\alpha_u} + \frac{p_{EE}^* - p_P^*}{2A_P|_E} dy \right) \\ &\quad - \frac{\alpha_u}{2} \left(\frac{1}{A_P|_P} + \frac{1}{A_P|_E} \right) (p_E^* - p_P^*) dy \\ &= \frac{1}{2} (u_P^*|_P + u_P^*|_E) - \frac{1}{2} \left(\frac{\alpha_u dy}{A_P|_P} + \frac{\alpha_u dy}{A_P|_E} \right) (p_E^* - p_P^*) \\ &\quad + \frac{1}{4} \frac{\alpha_u dy}{A_P|_P} (p_E^* - p_W^*) + \frac{1}{4} \frac{\alpha_u dy}{A_P|_E} (p_{EE}^* - p_P^*) \end{aligned} \quad (38)$$

4. Solve the pressure correction equation:
Defining

$$u_P^{(n)} := \alpha_u u_P + (1 - \alpha_u) u_P^{(n-1)} \quad (39)$$

and substituting into (24) gives

$$\frac{A_P}{\alpha_u} u_P^{(n)} = \sum_{nb} A_{nb} u_{nb} - \frac{p_E - p_W}{2} dy + S_u + \frac{A_P(1 - \alpha_u)}{\alpha_u} u_P^{(n-1)} \quad (40)$$

Subtracting (40) from (30) yields the momentum equation for the corrective terms

$$\frac{A_P}{\alpha_u} u_P^{(n)'} = \sum_{nb} A_{nb} u_{nb}' - \frac{p'_E - p'_W}{2} dy \quad (41)$$

The SIMPLE method drops the neighboring terms, yielding

$$\frac{A_P}{\alpha_u} u_P^{(n)'} = \frac{p'_W - p'_E}{2} dy \quad (42)$$

$$\Rightarrow u_P^{(n)'} = \alpha_u \frac{dy}{A_P} \frac{p'_W - p'_E}{2} \quad (43)$$

Defining $u_P' := u_P^{(n)} - u_P^{*(n)}$,

$$u_P^{(n)} = u_P^{*(n)} + \alpha_u \frac{dy}{A_P} \frac{p'_W - p'_E}{2} = u_P^{*(n)} + \frac{d}{2} (p'_W - p'_E) \quad (44)$$

where

$$d := \alpha_u \frac{dy}{A_P} \quad (45)$$

We do the same for the east face velocity u_e^* . The corrected face velocity is given by

$$u_P|_e = \frac{\alpha_u}{2} \left(\frac{u_P|_P}{\alpha_u} + \frac{p_E - p_W}{2A_P|_P} dy + \frac{u_P|_E}{\alpha_u} + \frac{p_{EE} - p_P}{2A_P|_E} dy \right) - \frac{\alpha_u}{2} \left(\frac{1}{A_P|_P} + \frac{1}{A_P|_E} \right) (p_E - p_P) dy \quad (46)$$

$$(47)$$

and subtracting out (38) gives the face velocity correction

$$\begin{aligned} u_P'|_e &= \frac{\alpha_u}{2} \left(\frac{u_P'|_P}{\alpha_u} + \frac{p'_E - p'_W}{2A_P|_P} dy + \frac{u_P'|_E}{\alpha_u} + \frac{p'_{EE} - p'_P}{2A_P|_E} dy \right) \\ &\quad - \frac{\alpha_u}{2} \left(\frac{1}{A_P|_P} + \frac{1}{A_P|_E} \right) (p'_E - p'_P) dy \\ &= \frac{\alpha_u}{2} \left(\frac{dy}{A_P} \frac{p'_W - p'_E}{2} + \frac{p'_E - p'_W}{2A_P|_P} dy + \frac{dy}{A_P} \frac{p'_P - p'_{EE}}{2} + \frac{p'_{EE} - p'_P}{2A_P|_E} dy \right) \\ &\quad - \frac{\alpha_u}{2} \left(\frac{1}{A_P|_P} + \frac{1}{A_P|_E} \right) (p'_E - p'_P) dy \\ &= -\frac{\alpha_u}{2} \left(\frac{1}{A_P|_P} + \frac{1}{A_P|_E} \right) (p'_E - p'_P) dy \\ &= d|_e (p'_P - p'_E) \end{aligned} \quad (48)$$

Likewise

$$v_P'|_n = d|_n (p'_P - p'_N) \quad (49)$$

where d is interpolated at the face using Rhie-Chow (37). Substituting (48) into the discretized continuity equation (12) to satisfy mass continuity yields

$$\begin{aligned} 0 &= \rho dy(u_P^*|_e + d|_e(p'_P - p'_E)) - \rho dy(u_P^*|_w + d|_w(p'_W - p'_P)) \\ &\quad + \rho dx(v_P^*|_n + d|_n(p'_P - p'_N)) - \rho dx(v_P^*|_s + d|_s(p'_S - p'_P)) \\ &\implies a_P p'_P = \sum_{nb} a_{nb} p'_{nb} + b \end{aligned} \quad (50)$$

where

$$\begin{aligned} a_e &= \rho d|_e dy, a_e = \rho d|_w dy, a_n = \rho d|_n dx, a_s = \rho d|_s dx \\ b &= \rho dy u_P^*|_e - \rho dy u_P^{*(n)}|_w + \rho dx u_P^*|_n - \rho dx u_P^*|_s \\ a_P &= \sum_{nb} a_{nb} \end{aligned} \quad (51)$$

Solve (50) to yield the corrective pressure field p' .

5. Update the pressure velocity fields for the cell centers and faces:
From (44)

$$u_P \leftarrow u_P + \frac{d}{2}(p'_W - p'_E) \quad (52)$$

$$v_P \leftarrow v_P + \frac{d}{2}(p'_S - p'_N) \quad (53)$$

From (48)

$$u_P|_e \leftarrow u_P|_e + d_e(p'_P - p'_E) \quad (54)$$

$$v_P|_n \leftarrow v_P|_n + d_n(p'_P - p'_N) \quad (55)$$

$$p_P \leftarrow p_P + \alpha_P p' \quad (56)$$

6. Check for convergence of u, v, p . If they are not yet converged, return to step 1 using the new fields as u^*, v^*, p^* . The Mean Absolute Error (MAE) was chosen to check for convergence:

$$|u - u^{(n-1)}| + |v - v^{(n-1)}| + |p'| < \epsilon * h^p \quad (57)$$

where h is the horizontal grid spacing and p is the accuracy order (2 for the central difference, 1 for the upwind scheme, and 1.5 for the hybrid scheme).

4 Results and Discussion

The preceding section outlines the general methodology of the code presented in this paper. The code can also be used as a solver for any arbitrary primitive, such as heat, though only solutions to problems involving laminar steady state flow are discussed in this section. All experiments are completed with $\alpha_u = \alpha_v = 0.7, \alpha_p = 0.3$.

4.1 Lid Driven Cavity

We test the solver on the classical Lid-Driven-Cavity problem; a square domain with no-slip walls and a lid on the top moving from left to right.

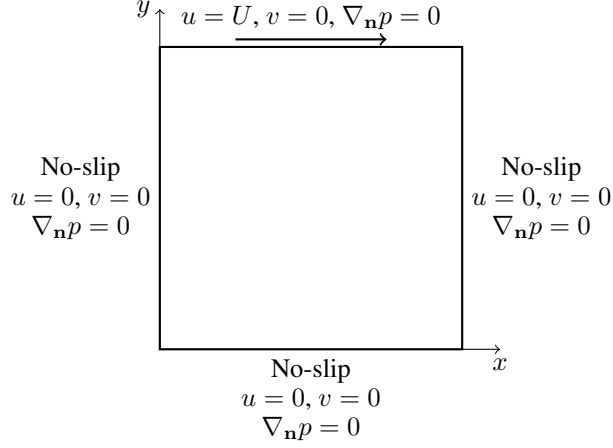


Figure 1: Lid-driven cavity setup. The top boundary is a moving lid with horizontal velocity U , while the left, right, and bottom boundaries are stationary no-slip walls. The domain is a square of size $L \times L$.

A grid size of 129×129 is used to compare with [1].

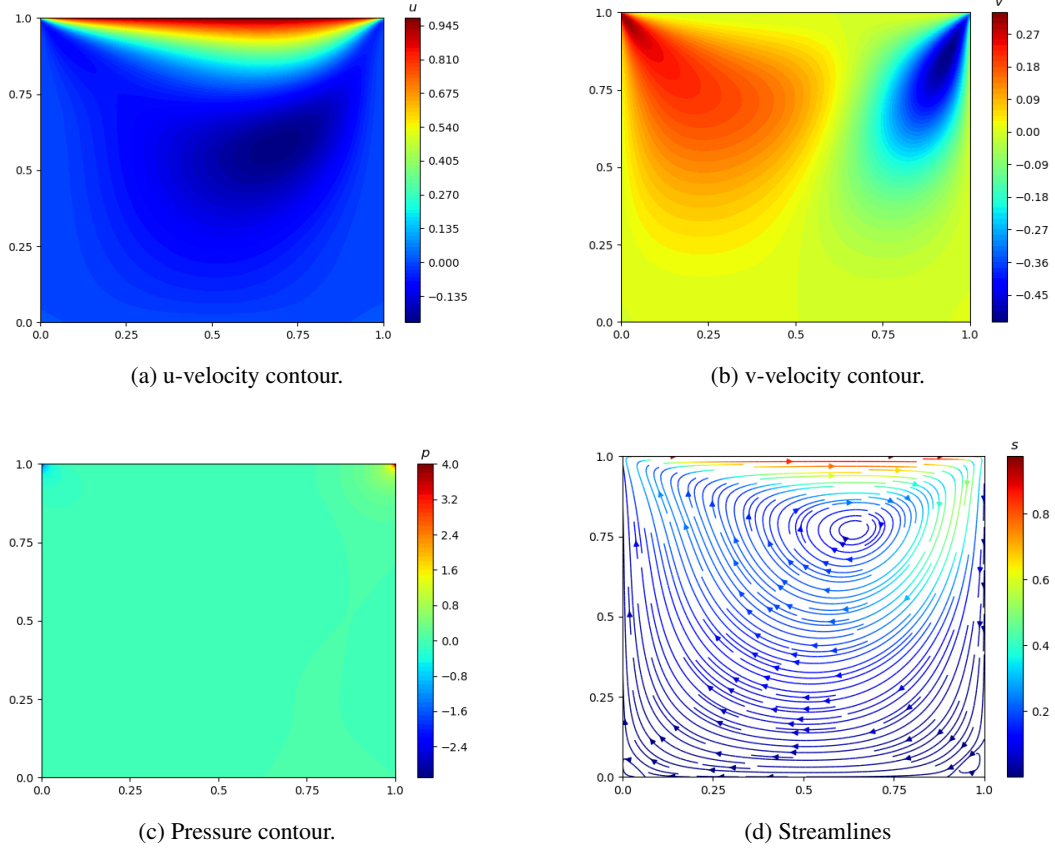


Figure 2: Lid-driven cavity results ($Re=100$, 129×129 grid): (a) Horizontal velocity component, (b) Vertical velocity component, (c) Pressure distribution, (d) Flow streamlines

The streamlines in 2 show one main vortex at around $(0.75, 0.75)$ and two minor vortices at the bottom corners of the domain. The velocity at the stationary walls is also near 0 highlighting the effect of the no-slip condition.

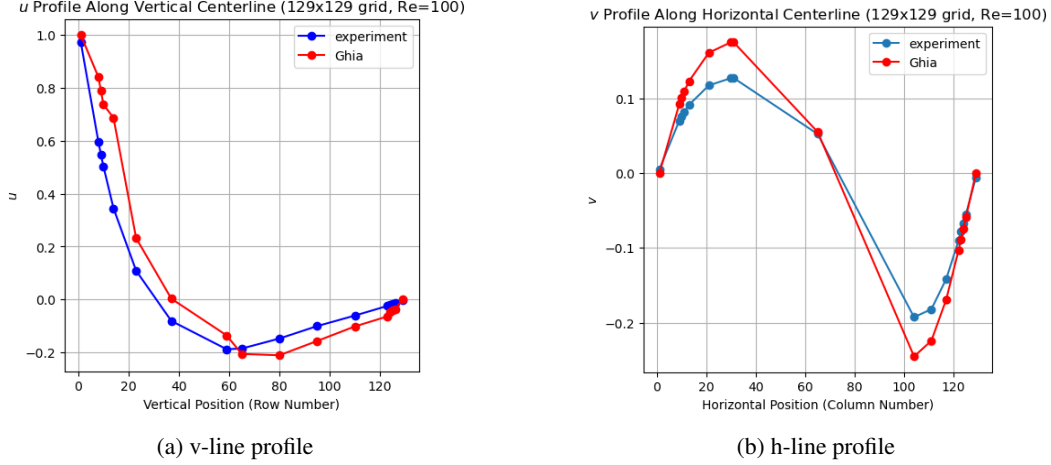


Figure 3: Velocity profiles along vertical (a) and horizontal (b) centerlines, comparing with [1]

There is sufficient agreement between figure 3 and [1], and the residuals likely result from underconvergence.

4.2 Lid Driven Cavity with Backward Facing Step

The lid driven cavity with a step at the lower left corner is solved next. The setup is presented as follows:

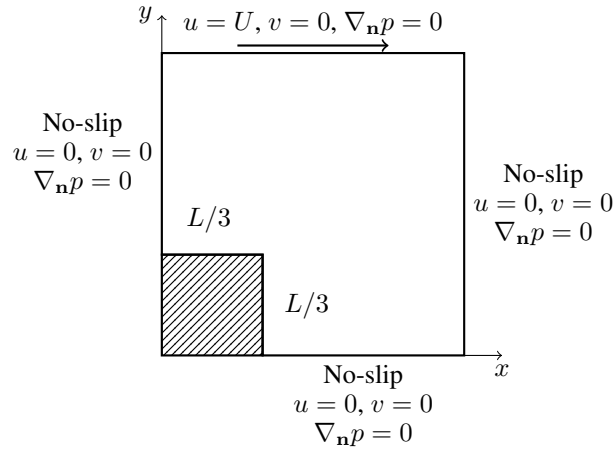


Figure 4: Lid-driven cavity with a backward-facing step. The domain includes a $L/3 \times L/3$ step at the bottom left (occupying $[0, L/3] \times [0, L/3]$). The top boundary is a moving lid with horizontal velocity U , while all other boundaries (including the step surfaces) are stationary no-slip walls. The pressure gradient across the backward facing step faces is also zero.

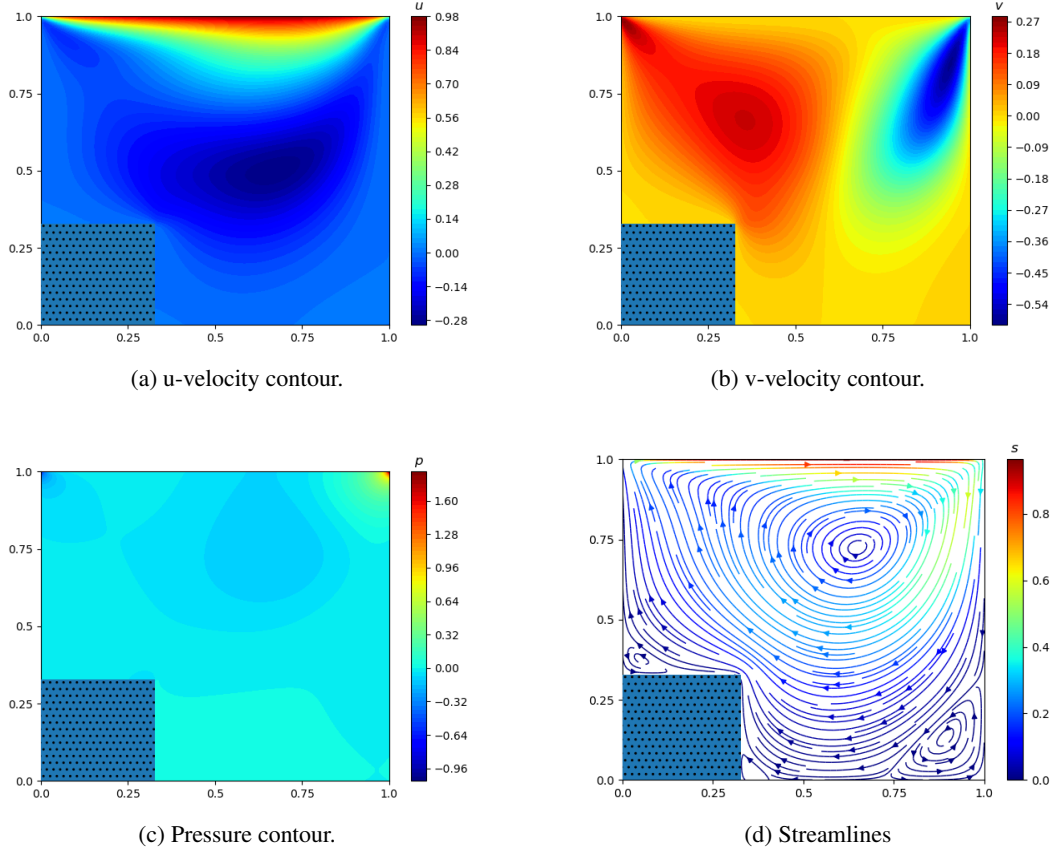


Figure 5: Lid-driven cavity with step results ($Re=200$, 100×100 grid): (a) Horizontal velocity component, (b) Vertical velocity component, (c) Pressure distribution, (d) Flow streamlines

The results in 5 show the step's effect on the flow around it, due to the no-slip conditions of its faces. The main vortex is present, as well as now a secondary vortex and two minor vortices; the secondary vortex at the lower right is analogous to the one observed in 2 and the two minor vortices are seen outside those corners of the step that are adjacent to a wall. The secondary vortex is larger than the one in 2 due to the increased Re and thus lower viscous forces driving the circulation.

4.3 Backward Facing Step

The code is then tested on the backward facing step problem, with a step at the lower left of the domain. The non-obstructed part of the left face is an inlet with flow perpendicular to the face and the right face is an outlet.

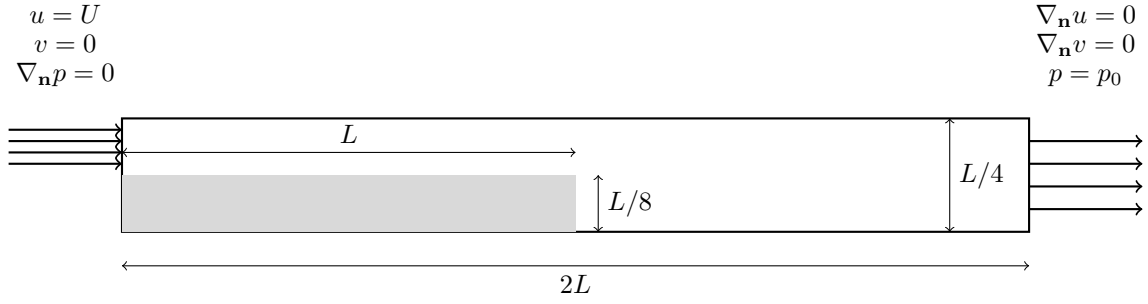
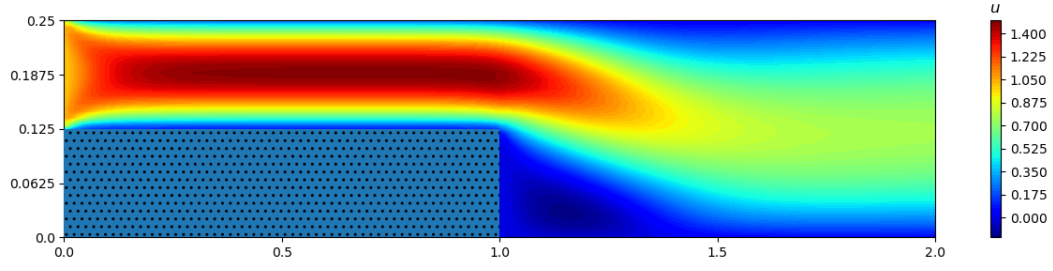
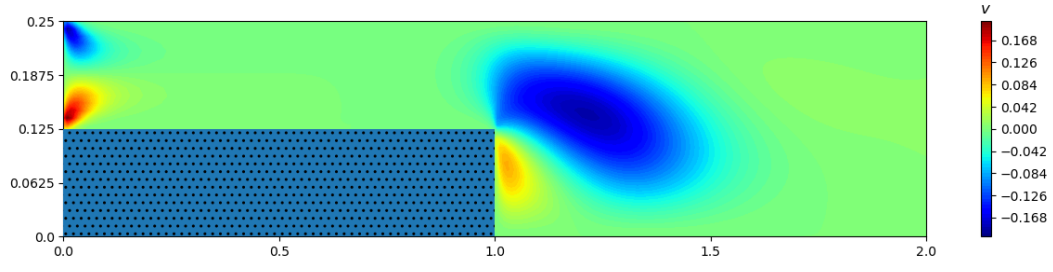


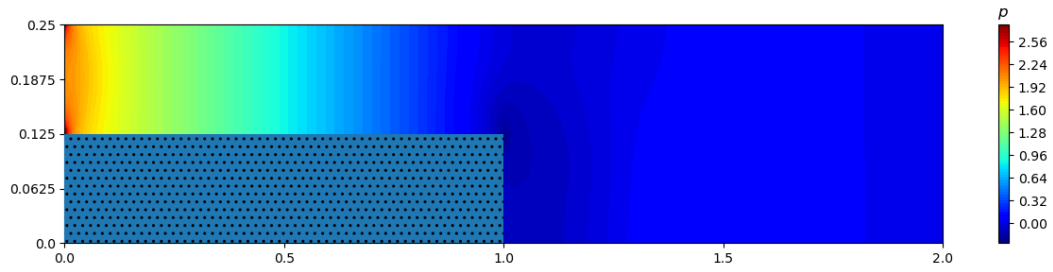
Figure 6: Schematic of the backward facing step problem. The domain has a width of $2L$ and a height of $L/4$. An obstacle of width L and height $L/8$ is placed at the lower left. The left inlet has an initial velocity U , while the right outlet has an unknown velocity.



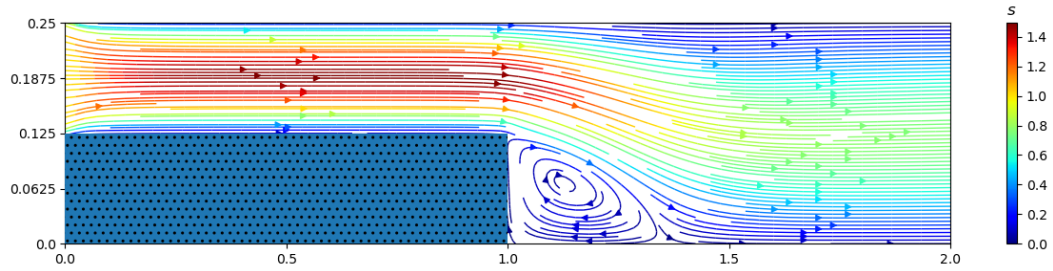
(a) u-velocity contour



(b) v-velocity contour

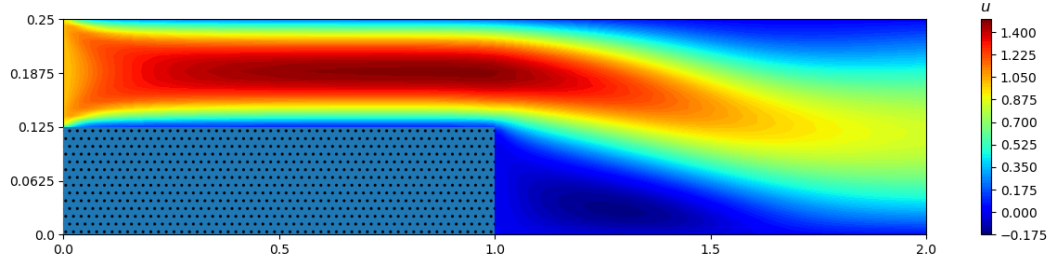


(c) Pressure contour

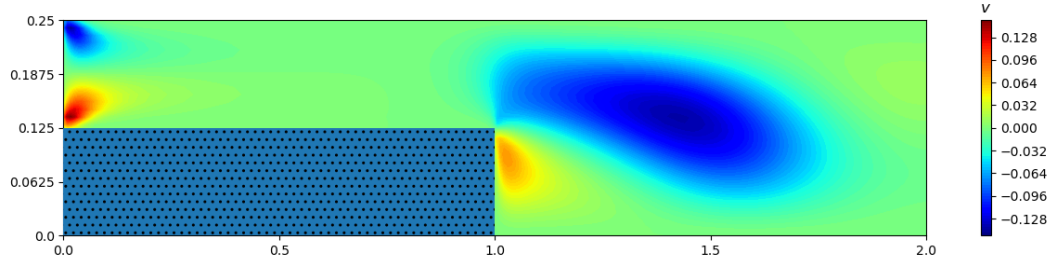


(d) Streamlines

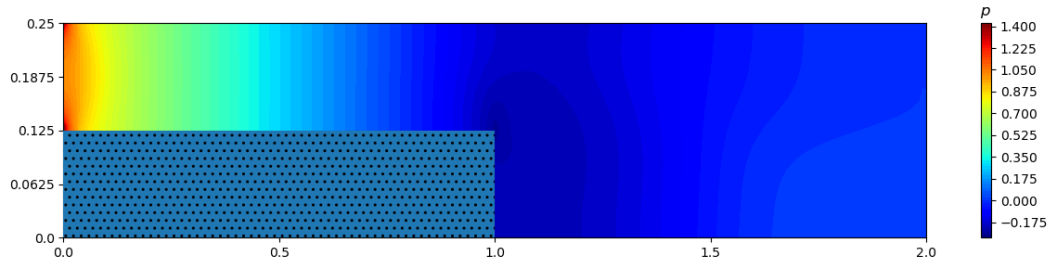
Figure 7: $Re = 100$ Backflow step solution on 30×240 grid.



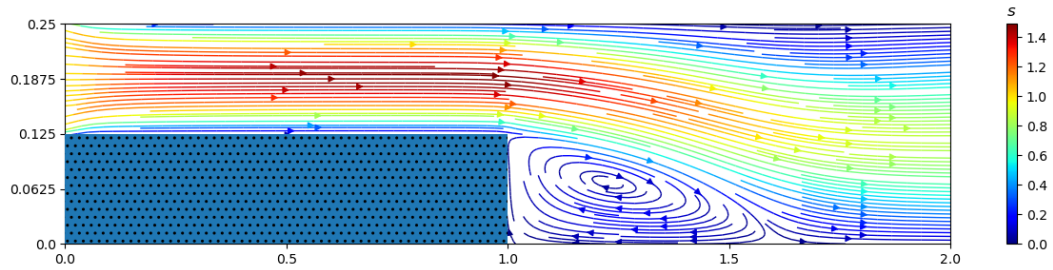
(a) u-velocity contour



(b) v-velocity contour



(c) Pressure contour



(d) Streamlines

Figure 8: $Re = 200$ Backflow step solution on 30×240 grid.

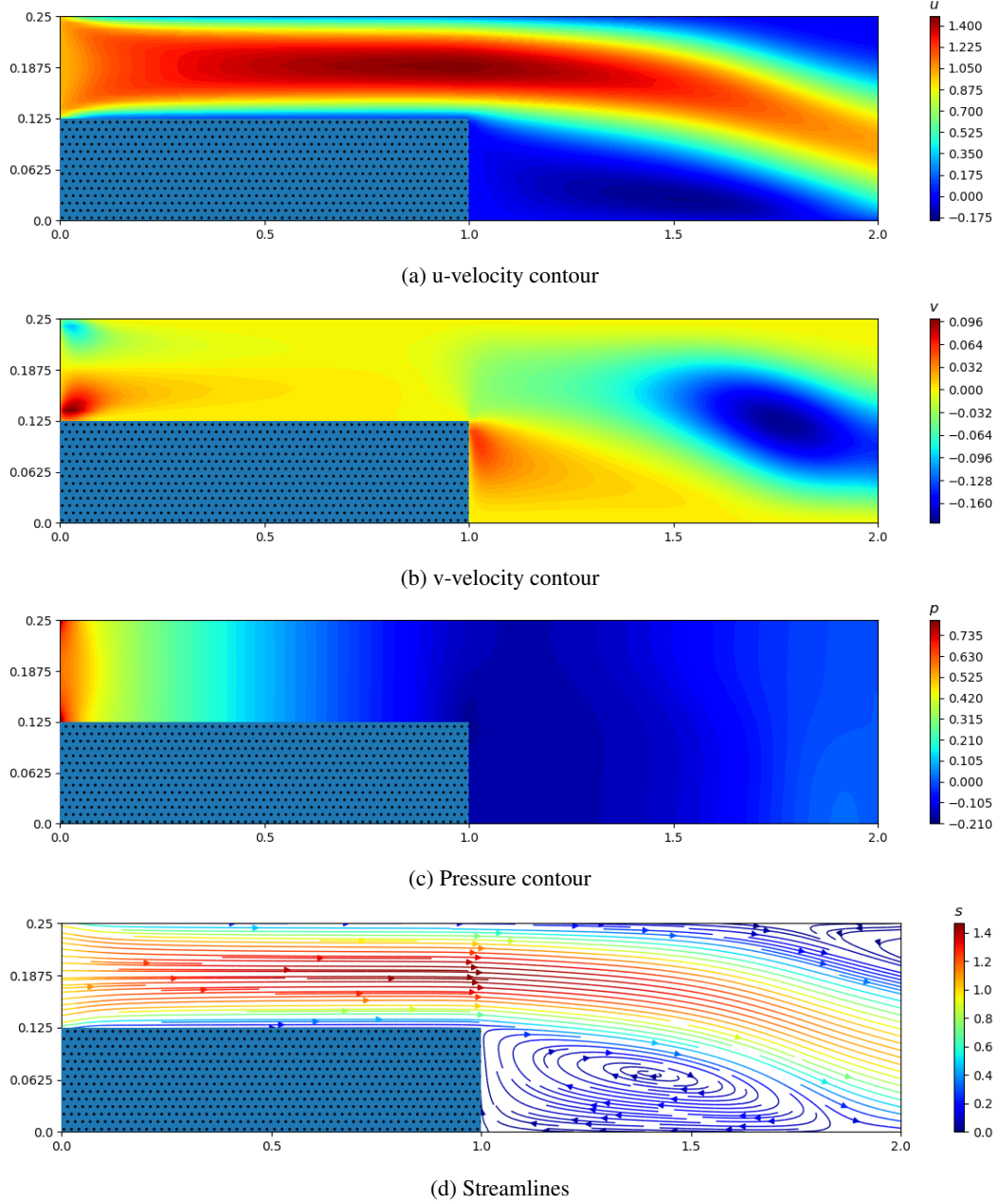


Figure 9: $Re = 400$ Backflow step solution on 30×240 grid.

Figures (7)-(9) show the increasing vortex size as the Reynolds number increases. Since u, L are kept constant this is equivalent to decreasing the kinematic viscosity. For laminar flow this is the expected result; low Re and thus stronger viscous forces leads to a small recirculation zone and reattachment length; as the viscous force decreases and Re increases, the reattachment length grows and the size of the recirculation zone increases. The no-slip condition is also observed by noting the velocity gradient heading to zero towards the top and bottom walls; while away from the walls the velocity increases to satisfy mass continuity. Numerical artifacts in the streamlines are also observed at the top of the outlet in 9, but since the flow is near zero, this is likely mitigated by improved convergence.

5 Conclusion and Code Capabilities

The code is able to solve key steady-state laminar flow problems in computational-fluid-dynamics problems including the lid-driven-cavity and the backward-facing-step problems. Vortices are observed and the effects of the Reynold's Number on the size of the vortices is noted. The effect of adding a step is also discussed, namely the additional no-slip walls causing the fluid to adhere to the surface with zero velocity as well as the presence of additional vortices. The code is written in an object-oriented manor, where the problem parameters such as domain dimensions, obstacle location, and boundary conditions are passed into a solver object which calls upon methods to generate a solution via the SIMPLE method. Multiple arrays are stored as fields such as an array for u, v, p as well as miscellaneous arrays to aid in locating cells of particular properties.

5.1 Boundary Conditions

The code allows the user to specify conditions for each of the boundaries. The available conditions include a Dirichlet condition and a homogeneous Neumann condition. Further work will need to be done to allow other types of boundary conditions such as the inhomogeneous Neumann condition and the Robin [mixed] condition.

5.2 Obstacles

The code also allows the placement of multiple obstacles at any position in the domain, though they are restricted to be rectangular in shape. They also must have homogeneous Neumann boundaries for any primitive including pressure, though they must have Dirichlet boundaries for the velocities to enforce the no-slip condition. Regarding the primitive boundary condition, this physically represents an insulation boundary, enforcing a zero gradient across.

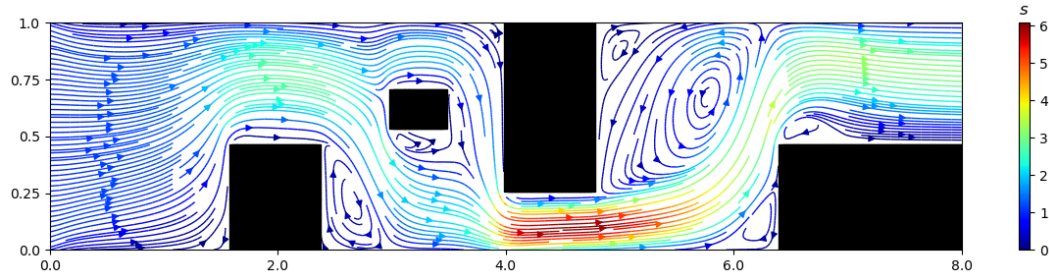


Figure 10: Inlet-outlet problem with many obstacles.

5.3 Future Work

In addition to the improvements in the code previously mentioned, more work will need to be done to allow for more diverse obstacle and domain shapes, perhaps for example a separate module facilitating the creation of arbitrary shapes. The code can also be further optimized to improve the convergence speed when solving for the velocity and pressure fields.

References

- [1] U. Ghia, K.N. Ghia, and C.T. Shin. “High-re solutions for incompressible flow using the navier-stokes equations and a multigrid method”. In: *Journal of Computational Physics* 48.3 (Dec. 1982), pp. 387–411. DOI: 10.1016/0021-9991(82)90058-4.
- [2] C. M. Rhie and W. L. Chow. “Numerical Study of the turbulent flow past an airfoil with trailing edge separation”. In: *AIAA Journal* 21.11 (Nov. 1983), pp. 1525–1532. DOI: 10.2514/3.8284.

A Grid Convergence Study

To show that the solution does not sufficiently change on grid refinement, a grid convergence study is conducted using a coarse resolution of 50×50 , a finer resolution of 100×100 , and a reference [exact] resolution of 150×150 . The results of the study done on the lid-driven cavity problem with a step is shown below:

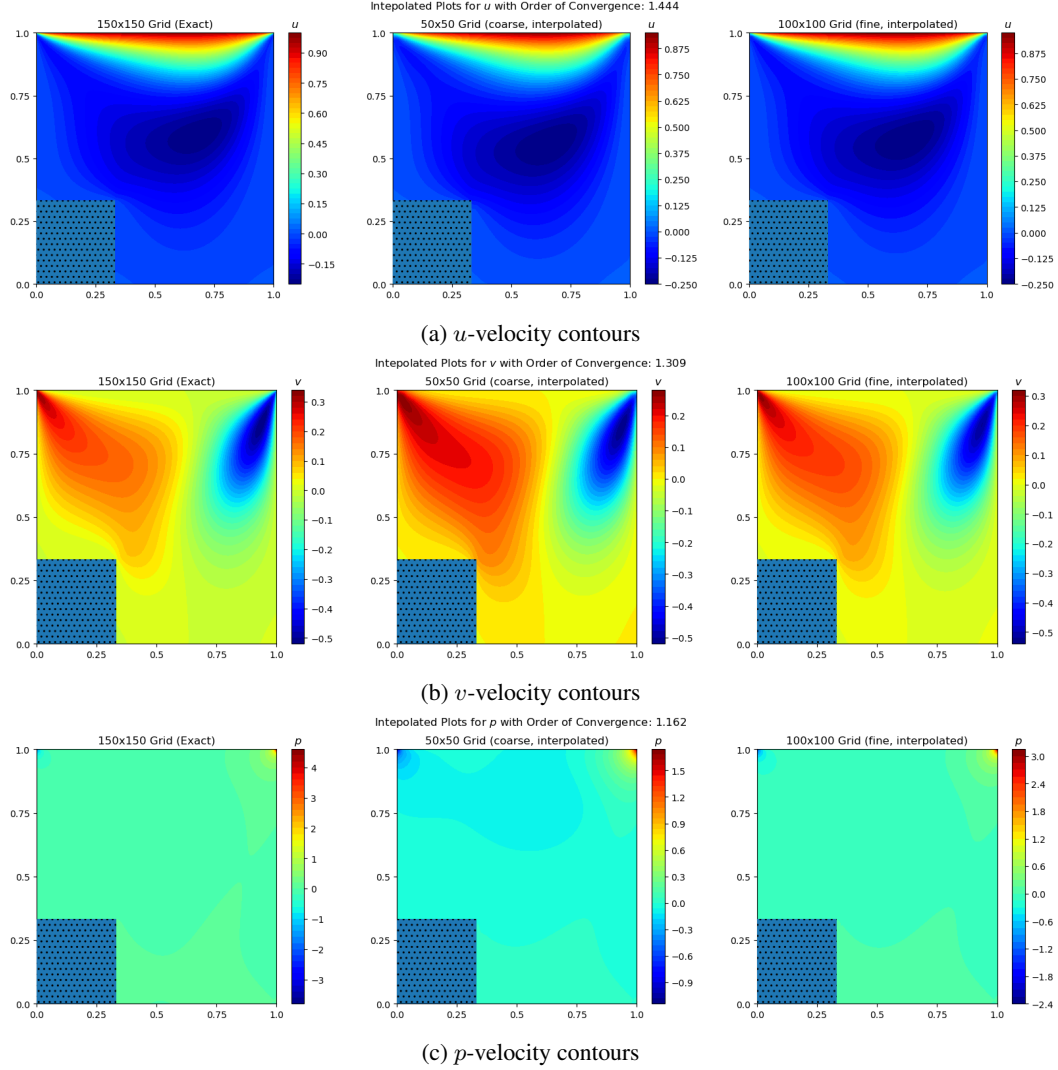


Figure 11: Grid convergence study, showing the interpolated plots in comparison to the reference 150×150 plots.